An Introduction to Functional Principal Component Analysis

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1 Introduction

1.1 Principal Component Analysis

In data analysis, *dimension reduction* is the practice of simplifying a complex dataset by reducing the number of variables. Oftentimes, dimension reduction techniques will trim away unimportant details leaving a dataset that is both more interpretable and has a smaller memory footprint. For these reasons, dimension reduction is often used as a first step in data exploration and compression. The earliest known method of dimension reduction is principal component analysis (PCA), which was introduced by Karl Pearson in 1901 [11]. The simplictly and effectiveness of PCA has seen it become the most widely used method of dimension reduction [8].

In the setting of PCA, we assume that observations $x^{(1)}, \dots, x^{(N)}$ are drawn from some multivariate distribution $X = (X_1, \dots, X_d) \in \mathbb{R}^d$. The observations are used to estimate the $d \times d$ population covariance matrix Σ, where $\Sigma_{ij} = \text{Cov}(X_i, X_j)$. Σ is positive semi-definite and thus admits an eigendecomposition

$$\Sigma = U \mathbf{\Lambda} U^T \tag{1.1}$$

where U is an orthonormal matrix of eigenvectors and $\mathbf{\Lambda}$ is a diagonal matrix of corresponding eigenvalues ordered from largest to smallest. The columns $u_1, \dots, u_d$ of U form a basis of $\mathbb{R}^d$, so we can write X as

$$X = \sum_{i=1}^{d} \langle X, u_i \rangle u_i \tag{1.2}$$

If $\mathbb{E}(X) = 0$, the orthogonality of the u_i implies that $\langle X, u_i \rangle$ and $\langle X, u_j \rangle$ are uncorrelated for $i \neq j$. The idea behind PCA is to approximate X by the first k terms in (1.2):

$$X \approx \sum_{i=1}^{k} \langle X, u_i \rangle u_i \tag{1.3}$$

This is the projection of X onto the k-dimensional subspace spanned by $u_1, \dots, u_k$. The reason for using the eigenvectors of Σ is that it provides (in a certain sense) the best k-dimensional approximation of X. In particular, [7] (section 7.2.3) shows that for any $1 \leq k \leq d$ and any orthogonal set of vectors $w_1, \dots, w_k \in \mathbb{R}^k$, we have

$$\mathbb{E}\big|\big|X - \sum_{i=1}^{k} \langle X, u_i \rangle u_i\big|\big|^2 \leq \mathbb{E}\big|\big|X - \sum_{i=1}^{k} \langle X, w_i \rangle w_i\big|\big|^2 \tag{1.4}$$

In other words, PCA finds a k-dimensional approximation of X that minimizes mean squared error.

1.2 Functional Data Analysis

Although PCA is applicable in a wide range of scenarios, it is not always the case that observed data is naturally represented by a vector in $\mathbb{R}^d$. [13] gives the example of the temperature in a city throughout a year. Although we collect measurements at only finitely many time points, the underlying process that we observe is a continuous function in time. If we are interested in N cities, the observed data are then N functions $f_1, \ldots, f_N$; see figure 1 for an example. Standard PCA cannot immediately be applied because the functional data are too granular to form a covariance matrix of finite size. Nevertheless, we may want investigate the dominant trends present in our data. Specifically, we might ask if the data can be well approximated by a linear combination of simpler functions $\phi_1, \ldots, \phi_k$:

$$f_i \approx \sum_{i=1}^{k} a_i \phi_i \tag{1.5}$$

The goal of **functional principal component analysis** (FPCA) is to find the best approximation of the form (1.5) (subject to some constraints on the ϕ_i).

In 1982, James Ramsay coined the phrase "functional data analysis (FDA)" to describe the setting of functional data [12]. In the same year, PCA was generalized to the FDA framework, although the techniques of FPCA can be found in literature as early as 1958 [5, 15]. In the 1990's and early 2000's FPCA was extended to more complicated scenarios such as sparse functional data and constraints on smoothness [17, 24]. Today, FPCA is considered one of the most important techniques in FDA [22]. FPCA has been used to analyze functional data in many fields including medicine, engineering, and finance [10, 21, 23].

The theoretical justification for FPCA is given by the 1946 theorem of Karhunen and Loève. This theorem and its implcations will be thoroughly covered in sections 2.1 and 2.2. Sections 2.3 and 2.4 will explain how FPCA is carried out in practice. Section 3 will discuss the strengths and weaknesses of the method. Finally, FPCA is demonstrated on both synthetic datasets and US stock price data in section 4.

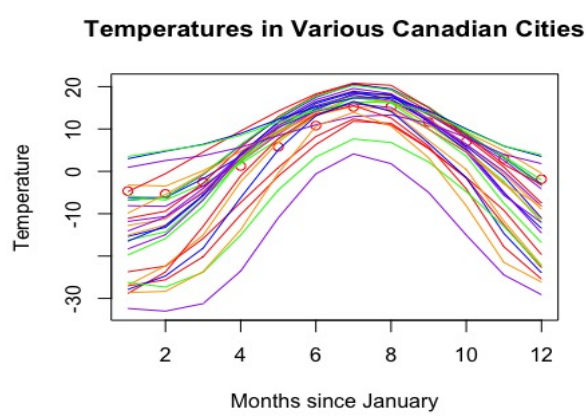


Figure 1: The temeperature curves (in celsius) for 35 Canadian cities. The data was obtained from the R package `fda` [14]. In functional data analysis, each curve is one data point. FPCA will provide a way to analyze variation between curves.

2 Methods

A stochastic process $X = \{X_t\}_{t\in[0,1]}$ is said to be **square-integrable** if $\sup_t \mathbb{E}(X_t^2) < \infty$. Define the **mean trajectory** $\mu(t) = \mathbb{E}(X_t)$. If $\mu(t) = 0$, then the process is **centered**. For a centered process, we define the map $K : [0,1] \times [0,1] \to \mathbb{R}$ to be the covariance between X_s and X_t:

$$K(s,t) = \mathrm{Cov}(X_s, X_t) = \mathbb{E}(X_s X_t) \tag{2.1}$$

In the setting of FPCA, we assume that the observed data $f^{(1)}, \ldots, f^{(N)}$ are independent realizations of a square-integrable process X. For convenience, we will also assume that X is centered (although this assumption can be relaxed).

2.1 The Karhunen-Loève Expansion

As mentioned earlier, we cannot form a covariance matrix for X because we cannot encode $\mathrm{Cov}(X_s, X_t)$ for all combinations of s and t. Instead, we define the **covariance operator**:

$$C : L^2([0,1]) \longrightarrow L^2([0,1]) \tag{2.2}$$

$$f \longmapsto \int_{[0,1]} K(s,t) f(t) dt \tag{2.3}$$

It is easy to see that C defines a linear operator on $L^2([0,1])$.[1] C is an immediate generalization of the covariance matrix Σ. When X is a finite dimensional distribution, we have

$$(\Sigma x)_i = \sum_{j=1}^{d} \mathrm{Cov}(X_i, X_j) x_j \tag{2.4}$$

The cornerstone of FPCA is a generalization of the eigendecomposition of Σ. To begin, we generalize the spectral theorem to $L^2([0,1])$. An operator H defined on $L^2([0,1])$ has eigenvalue λ if there exists a function f such that $H(f) = \lambda f$. The function f is called an **eigenfunction** of H. H is **self-adjoint** if $\langle H(f), g\rangle = \langle f, H(g)\rangle$ for all $f, g \in L^2([0,1])$. Notice that in finite dimensions, the operator $T : \mathbb{R}^d \to \mathbb{R}^d$ induced by a $d \times d$ symmetric matrix is self-adjoint. Then by the usual spectral theorem, there exists an orthonormal basis of $\mathbb{R}^d$ given by the eigenvectors of T. A similar result holds in arbitrary Hilbert spaces:

Theorem 1 (Spectral)**.** *Let $H : L^2([0,1]) \to L^2([0,1])$ be a compact*[2] *and self-adjoint operator. Then there exists an orthonormal basis of $L^2([0,1])$ given by the eigenfunctions of H. Furthermore, the corresponding eigenvalues are real.*

Notice that there are countably many eigenfunctions of H. The above theorem implies that any element in $L^2([0,1])$ can be written as the countable sum of these eigenfunctions.

The covariance operator C as defined in (2.3) satisfies the conditions of the spectral theorem. A simple application of Fubini's theorem shows that C is self-adjoint. The fact that C is compact is a

[1] To show that $C(f) \in L^2([0,1])$, apply Cauchy-Schwarz

[2] An operator is **compact** if the image of the open unit ball is relatively compact. That is, its closure is compact. See [1] for details.

result of the theory of Hilbert-Schmidt operators; see [1] (section 2.1) for details. The eigenvalues of C are in fact non-negative. C is a **positive operator**, which means that $\langle C(f), f\rangle \geq 0$ for any f. Positive operators always have non-negative eigenvalues. If f is an eigenfunction of C, then $\langle f, C(f)\rangle = \lambda||f||^2$, which forces $\lambda \geq 0$.

The following famous result summarizes the above findings.

Theorem 2 (Karhunen-Loève)**.** *Let X be square-integrable and centered. Then*

$$X = \sum_{i=1}^{\infty} \langle X, \phi_i\rangle \phi_i \tag{2.5}$$

where ϕ_i are eigenfunctions of the covariance operator C (as defined in (2.3)). Moreover,

1. $\mathbb{E}(\langle X, \phi_i\rangle) = 0$
2. $\mathbb{E}(\langle X, \phi_i\rangle\langle X, \phi_j\rangle) = 0$ *if* $i \neq j$
3. $\mathbb{E}(\langle X, \phi_i\rangle^2) = \lambda_i$, *where λ_i is the eigenvalue corresponding to ϕ_i.*

We call (2.5) the Karhunen-Loève expansion (KLE) of X. A proof of theorem 2 begins by applying the spectral theorem to C to obtain the ϕ_i. Items 1, 2, and 3 are simple computations relying on the fact that the eigenfunctions are orthogonal. Completing the proof requires showing that the sum in (2.5) converges to X in L^2. This argument requires Mercer's theorem and can be found in [1].

2.2 Optimality of the KLE

Assume that we have decomposed X as in (2.5) such that the eigenvalues corresponding ϕ_i are in decreasing order. Similar to PCA, we can approximate X by the first k terms in the infinite sum:

$$\hat{X} = \sum_{i=1}^{k} \langle X, \phi_i\rangle \phi_i \approx X \tag{2.6}$$

In fact, the approximation in (2.6) is optimal in the sense that it minimizes the mean squared error of the approximation.

Theorem 3. *Let $\{\psi_i\}_{i=1}^{\infty}$ be any orthonormal basis of $L^2([0,1])$. Then*

$$\mathbb{E}||X - \hat{X}||^2 \leq \mathbb{E}||X - \sum_{i=1}^{k} \langle X, \psi_i\rangle \psi_i||^2 \tag{2.7}$$

See appendix A for a proof of theorem 3.

The KLE can also be interpreted as providing k orthonormal functions that best captures the variance of the process X. Since X is square-integrable,

$$\int_{[0,1]} \mathrm{Var}(X_t)dt = \int_{[0,1]} \mathbb{E}X_t^2 dt < \infty \tag{2.8}$$

Because $\langle X, \phi_i \rangle$ and $\langle X, \phi_j \rangle$ are uncorrelated for $i \neq j$, (2.8) implies

$$\int_{[0,1]} \mathrm{Var}(X_t) dt = \int_{[0,1]} \sum_{i=1}^{\infty} \mathbb{E}\left(\langle X, \phi_i \rangle \phi_i(t)\right)^2 dt \tag{2.9}$$

$$= \int_{[0,1]} \sum_{i=1}^{\infty} \phi_i^2(t) \lambda_i dt \tag{2.10}$$

$$= \sum_{i=1}^{\infty} \lambda_i \tag{2.11}$$

In particular, $\sum_{i=1}^{\infty} \lambda_i < \infty$. We can thus measure the fraction of variance explained by k basis functions by computing

$$\frac{\int_{[0,1]} \mathrm{Var}\left(\sum_{i=1}^{k} \langle X, \phi_i \rangle \phi_i(t)\right) dt}{\int_{[0,1]} \mathrm{Var}(X_t) dt} = \frac{\sum_{i=1}^{k} \lambda_i}{\sum_{i=1}^{\infty} \lambda_i} \tag{2.12}$$

If the proportion in (2.12) is close to 1, then the k basis functions are a good approximation for X. It turns out that the first k basis functions in the KLE of X are optimal in the sense of maximizing (2.12).

Theorem 4. *If $\{\psi_i\}_{i=1}^{\infty}$ is any orthonormal basis of $L^2([0,1])$, then*

$$\sum_{i=1}^{k} \lambda_i \geq \int_{[0,1]} Var\left(\sum_{i=1}^{k} \langle X, \psi_i \rangle \psi_i(t)\right) dt \tag{2.13}$$

for any k.

Proof. By theorem 3,

$$\sum_{i=k+1}^{\infty} \lambda_i \leq \int_{[0,1]} \mathbb{E}\left(\sum_{i=k+1}^{\infty} \langle X, \psi_i \rangle \psi_i(t)\right)^2 dt = \int_{[0,1]} \mathrm{Var}\left(\sum_{i=k+1}^{\infty} \langle X, \psi_i \rangle \psi_i(t)\right) dt \tag{2.14}$$

Therefore,

$$\sum_{i=1}^{k} \lambda_i = \sum_{i=1}^{\infty} \lambda_i - \sum_{i=k+1}^{\infty} \lambda_i \tag{2.15}$$

$$\geq \sum_{i=1}^{\infty} \lambda_i - \int_{[0,1]} \mathrm{Var}\left(\sum_{i=1}^{k+1} \langle X, \psi \rangle \psi_i(t)\right) dt \tag{2.16}$$

$$= \int_{[0,1]} \mathrm{Var}\left(\sum_{i=1}^{k} \langle X, \psi_i \rangle \psi_i(t)\right) dt \tag{2.17}$$

□

Returning to the setting of functional data $f_1, \ldots, f_N$ drawn from X, we can express each f_i in terms of the truncated KLE basis:

$$f_i \approx \sum_{i=1}^{k} \langle f_i, \phi_i \rangle \phi_i \tag{2.18}$$

This approximation has been shown to be optimal in several senses, so we have essentially solved the problem posed in (1.5). As a tribute to PCA, we call $\langle f_j, \phi_i \rangle$ the **i-th functional principal component (FPC) score** of f_j.

2.3 Estimation

In practice, we can only collect finitely many measurements from each f_i. Consequently, the covariance operator C must be estimated using interpolation techniques. In the simplest case, all f_i are observed at the *same* time points $t_1 \leq \ldots \leq t_m$. In other words, we have $f_i(t_j)$ for $1 \leq i \leq N$ and $1 \leq j \leq m$. In this case, we can estimate the mean trajectory μ and covariance K as:

$$\hat{\mu}(t_j) = \frac{1}{N} \sum_{i=1}^{N} f_i(t_j) \tag{2.19}$$

$$\hat{K}(t_j, t_k) = \frac{1}{N} \sum_{i=1}^{N} (f_i(t_j) - \hat{\mu}(t_j))(f_i(t_k) - \hat{\mu}(t_k)) \tag{2.20}$$

Then we can estimate the covariance operator at any time point t_i as

$$\hat{C} f(t_i) = \sum_{j=1}^{m} \hat{K}(t_j, t_i) f(t_j) \tag{2.21}$$

To estimate the eigenfunctions of C, we need to find f satisfying

$$\sum_{j=1}^{m} \hat{K}(t_i, t_j) f(t_j) = \lambda f(t_i) \quad \text{for } i = 1, \ldots, m \tag{2.22}$$

If we express $\hat{C}$ as a $m \times m$ matrix where $\hat{C}_{ij} = \hat{K}(t_i, t_j)$, then (2.22) can be rephrased as finding $v \in \mathbb{R}^m$ such that

$$\hat{C} v = \lambda v \tag{2.23}$$

$\hat{C}$ is a (positive semi-definite) sample covariance matrix, so v and λ can be found by computing the eigendecomposition of $\hat{C}$. In particular, the k-th eigenfunction of C can be estimated by $\hat{\phi}_k(t_i) = v_{i,k}$, where $v_{i,k}$ is the i-th component of the eigenvector of $\hat{C}$ corresponding to the k-th largest eigenvalue. We can plot the eigenfunction estimate by interpolating between the computed values of $\hat{\phi}_k$. [13] notes that any method of interpolation can provide accurate estimates provided the time points are sufficently close together. Indeed, [22] provides mild conditions for which $\hat{\phi}_k$ converges to the true eigenfunction ϕ_k. Despite the possibility of infinite non-zero eigenvalues in the KLE, only N estimated eigenfunctions can have non-zero eigenvalue as the rank of $\hat{C}$ is bounded by N. In practice, however, this constraint is unlikely to be problematic as we are usually concerned with only the first few eigenfunctions.

The above procedure can be summarized in terms of standard PCA. Given N functions observed at time points $t_1 \leq \ldots \leq t_m$ form the $N \times m$ matrix X, where $X_{ij} = f_i(t_j)$. Using any statistical software[3], conduct PCA on X. The eigenfunction estimates $\hat{\phi}_k$ are given by interpolating the principal component loadings. The returned principal component scores can be used as estimates of the FPC scores.

[3] For example, R's `prcomp` routine.

A more difficult scenario is when the observations are sparse and irregularly sampled. In this situation, we can only obtain a few measurements from each f_i at times that vary with i. This is often the case in longitudinal studies where participants might be examined at varying times. When there are only a few observations at each time point, the estimators $\hat{\mu}$ and $\hat{K}$ provided in (2.19) and (2.20) are highly sensitive to noise and thus will not give an accurate representation of the underlying process. Moreover, the FPC scores $\langle \phi_i, f_j \rangle$ can not be reliably estimated by $\sum_t \phi_i(t) f_j(t)$ due to an insufficient number of observations.

There have been a variety of methods designed to overcome this problem [16, 18, 20]. Yao et al. proposed Principal Component Analysis through Conditional Expectation (PACE), which uses local linear smoothing to estimate μ and K [24]. Once a smooth estimate for K is obtained, $\hat{C}$ can be computed as in (2.21) by discretizing K to an appropriately spaced grid. The FPC scores are estimated by computing the expected value of $\int \phi_i(t) f(t) dt$ conditional on the observed values of f_j, see [24] (section 2.3) for details.

2.4 Regularization

There is no guarantee that the eigenfunctions returned by FPCA will be well behaved. In many cases, however, it is important that the basis functions are simple and interpretable. **Regularized FPCA** solves this problem by penalizing noisy basis functions. There are many ways that this can be accomplished, but our exposition will focus only on the method given in [13] (chapter 7).

Let D denote the derivative operator. In this section, we are only interested in functions with square-integrable second derivative. In other words, functions f such that $||D^2(f)||_{L^2}^2 < \infty$. We can define the **roughness** of f as

$$R(f) = ||D^2(f)||^2 \tag{2.24}$$

If f is very noisy, then f will likely have many inflection points and $R(f)$ will be large.

For the moment we will focus on estimating the eigenfunction ϕ_1 corresponding to the largest eigenvalue λ_1. Recall that the optimization can be formulated as

$$\phi_1 = \max_{||f||=1} \int \mathrm{Var}\left(\langle X, f \rangle f(t)\right) dt \tag{2.25}$$

Equivalently, we could divide by $||f||^2$ and optimize over any ϕ:

$$\phi_1 = \max_f \frac{\int \mathrm{Var}(\langle X, f \rangle f(t)) dt}{||f||^2} = \frac{\mathrm{Var}(\langle X, f \rangle) \int f^2(t)}{||f||^2} \tag{2.26}$$

Of course, we have shown that (2.26) is maximized by choosing f to be the eigenfunction of C corresponding to λ_1. The objective can thus be expressed as

$$\phi_1 = \max_f \frac{\langle f, C(f) \rangle}{||f||^2} \tag{2.27}$$

To penalize noisy functions, we incorporate the roughness term and obtain the new objective

$$\max_f \frac{\langle f, C(f) \rangle}{||f||^2 + \rho R(f)} \tag{2.28}$$

where $\rho > 0$ is a tuning parameter that controls the amount of roughness that will be tolerated.

To maximize (2.28), we first fix smooth orthonormal basis functions $\{\psi_i\}_{i=1}^{\infty}$ over $[0,1]$ (or any closed interval). Let $c_i \in \mathbb{R}^k$ be the coefficients of f_i in the first k basis functions. Let $\hat{C}$ be the covariance matrix obtained from the data matrix with rows $c_1, \ldots, c_N$. Finally, let K be a $k \times k$ matrix such that $K_{ij} = \langle D^2\psi_i, D^2\psi_j \rangle$. Then the discrete approximation of (2.28) is

$$\max_{x \in \mathbb{R}^k} \frac{x^T \hat{C} x}{x^T x + \rho x^T K x} = \max_{x \in \mathbb{R}^k} \frac{x^T \hat{C} x}{x^T (I + \rho K) x} \tag{2.29}$$

Since ρ is positive, $A = I + \rho K$ is positive definite and is hence invertible. If we set $y = A^{-1/2}x$, we can rewrite (2.29) as

$$\max_{y \in \mathbb{R}^k} \frac{y^T A^{-1/2} \hat{C} A^{-1/2} y}{y^T y} \tag{2.30}$$

This is maximized by choosing y to be the eigenvector corresponding to the largest eigenvalue of $A^{-1/2}\hat{C}A^{-1/2}$. We thus obtain the eigenequation

$$\hat{C}x = \lambda(I + \rho K)x \tag{2.31}$$

Let $u_i \in \mathbb{R}^k$ be the vector corresponding to the i-th largest value of λ in (2.31). Then our estimate for the regularized eigenfunction ϕ_i is

$$\hat{\phi}_i(t) = \sum_{i=1}^{k} u_i \psi_i(t) \tag{2.32}$$

See section 7.4.2 of [13] for the details on how (2.31) can be solved in practice.

The choice of $\{\psi_i\}$ and k depends on the data. For example, if the functional data are periodic, then the Fourier basis given by

$$f_n(x) = \sqrt{2}\sin(2\pi n x) \qquad g_n(x) = \sqrt{2}\sin(2\pi n x) \tag{2.33}$$

for all $n \in \mathbb{N}$ is a good choice. In the non-periodic case, a set of orthogonal polynomials such as the Hermite polynomials might be a good choice[4]. The error (in L^2 norm) between f and its approximation by $\{\psi_i\}_{i=1}^{k}$ can be used to choose an appropriate value of k.

3 Discussion

3.1 Advantages over PCA

The main advantage of FPCA is that it provides a systematic way to analyze functional data. When the functional data are densely sampled FPCA can be computed by simply interpolating the eigenvectors of standard PCA. In more complicated sampling schedules, however, standard PCA is unable to accurately capture the underlying functional process. For example, in the case of irregularly sampled data, we saw that more sophisticated methods such as PACE are required to effectively estimate the mean trajectory and covariance.

Even though functional data is always discretized in practice, there is a lot to be gained by not viewing the data simply as a vector in $\mathbb{R}^d$. The theorem of Karhuhen and Loève extends the

[4] Although these are only orthogonal over $\mathbb{R}$.

eigendecomposition of a covariance matrix to infinite dimensional Hilbert spaces. We saw that this allows us to decompose the underlying stochastic process into finitely many orthogonal basis functions. Moreover, we saw that these basis functions were optimal in the sense of explained variance and mean squared error. The importance of these theoretical insights is that it tells us why the procedure of the smoothly interpolating the eigenvectors is more than just an ad-hoc method. In practice, the covariance operator is usually well approximated by the covariance matrix and the functional data are approximated by a high dimensional vector. The setting of functional data allows us to quantify this approximation error [22].

Regularized FPCA does not have a natural interpretation in terms of standard PCA. We defined roughness as the L^2 norm of the second derivative, which only makes sense in the setting of functional data. We incorporated the roughness penalty into the optimization objective of FPCA. If the data can be written as a linear combination of finitely many smooth basis functions, then this objective reduces to the eigenequation (2.31).

Standard PCA relies on an eigendecomposition of the $d \times d$ sample covariance matrix. When $d \gg N$, the sample covariance matrix is not a good estimator of the true population covariance matrix [19]. In the case where the functional data is smooth, the resulting eigenfunctions might be very noisy. In FPCA, one can smooth the covariance estimate $\hat{K}$ or apply regularized FPCA to obtain more interpretable basis functions.

Many of the strengths of FPCA are inherited from standard PCA. For example, **functional regression** models a stochastic process Y as an inner product of X and a coefficient function α [22]:

$$Y = \langle X, \alpha \rangle = \int X_t \alpha(t) dt \tag{3.1}$$

Similar to principal component regression, we can replace X by its truncated Karhunen-Loève transformation. Ideally, simplifying X reduces the difficultly of estimating α. This method is aptly named **functional principal component regression** [22].

Similar to PCA, FPCA can be used as a preliminary step in constructing parametric models [22]. By viewing the first few eigenfunctions, one can view the dominant curves and trends in the data. This analysis may elucidate a parametric model of the data generating process which can be used to better estimate and understand the functional data.

3.2 Shortcomings of FPCA

FPCA inherits many of the problems in standard PCA. Most notably, FPCA is restricted to projecting the functions onto a *linear* subspace spanned by the k basis functions. Similar to PCA, FPCA is unable to detect non-linear subspaces. For example, consider functional data generated on the unit ball in L^2, which can be written as the level set of a single equation:

$$\{f \in L^2 : \int f^2 = 1\} \tag{3.2}$$

However, FPCA will not find that a subspace of dimension 1 is sufficient for explaining all variance in the data. In finite dimensions, there are well known non-linear dimension reduction techniques such as ISOMAP [2]. Recently, there has been research extending these methods to functional data [3].

FPCA is also sensitive to outliers. The first basis function in FPCA represents the one dimensional subspace of L^2 explaining the maximum amount of variance in the data. If one function f^* is

significantly larger (in L^2 norm) than all other functions, then the objective of FPCA can easily be maximized by choosing $\phi_1 = f^*/||f^*||$. However, this ignores all of the other data, as f^* was chosen simply because its size. To prevent this, one can visually inspect the data and remove obvious outliers. There have been methods designed to make FPCA robust to outliers, although [22] claims that more research is needed in this area.

There are difficulties associated with assessing the performance of FPCA. [22] notes that as k (the number of eigenfunctions) increases, the amount of variance that can be explained increases. However, since each eigenfunction is approximated by discretizing the data, a larger k is associated with more random error. This introduces a bias-variance tradeoff in the choice of k, with the possibility of overfitting if k is large and N is small. Unfortunately, [22] claims that determining optimal value of k is still an open problem. Methods from standard PCA such as scree plots and (estimated) variance explained are still commonly used to asses performance despite ignoring the possibility of overfitting.

It may also be difficult to choose the value of ρ in the penalized objective (2.28). Perhaps the most common approach is to choose a value of ρ that gives desirable results through trial and error. Ramsay provides a cross validation procedure for choosing ρ in [13] (section 7.5). According to [22], there is no research supporting a theoretically optimal value of ρ.

Finally, there is no guarantee that the eigenfunctions or FPC scores have physical interpretations. As we will see in the following section, the eigenfunctions and FPC scores can be negative even when the outputs of the functional data are strictly positive. In finite dimensions, non-negative matrix factorization (NMF) [9] solves this problem by forcing the scores to be non-negative. It would thus be interesting to see if NMF can generalize to functional data.

4 Examples

4.1 Synthetic Data Experiments

First, we investigate FPCA in the case of densely sampled functional data. We generate 200 random curves in the Fourier basis (2.33) over $[0, 1]$ via the following procedure

$$f(x) = \sum_{k=1}^{25} a_k \sin(2\pi kx) + b_k \cos(2\pi kx) \tag{4.1}$$

where $a_k, b_k \sim \mathcal{N}(0, \frac{1}{k^2})$. We sample from each curve at the same x with an equal spacing of 0.001. As shown in figure 2, overlaying all 200 curves is not particularly enlightening. Once FPCA is applied, however, the underlying process becomes clear. Figure 3a shows the first four eigenfunction estimates. Notice that the first and second eigenfunctions correspond exactly to $\sin(2\pi x)$ and $\cos(2\pi x)$ and the third and fourth eigenfunctions correspond exactly to $\cos(4\pi x)$ and $\sin(4\pi x)$. In other words, FPCA identified the exact the basis functions that were used to generate the data. Despite theoretical concerns presented in the previous section, the simplicity of scree plots make them a good tool for analyzing the performance of FPCA. The scree plot in 3b indicates that the data can be completely explained by 50 basis functions, exactly the number of basis functions used in (4.1).

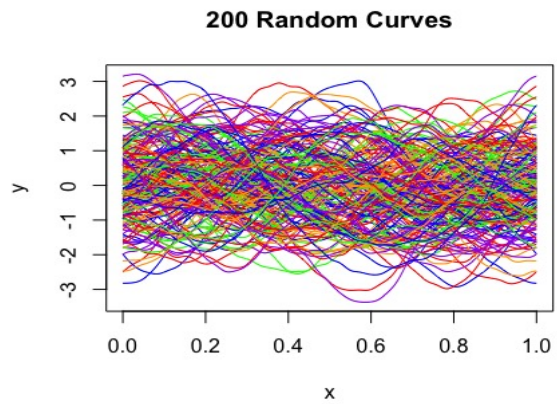
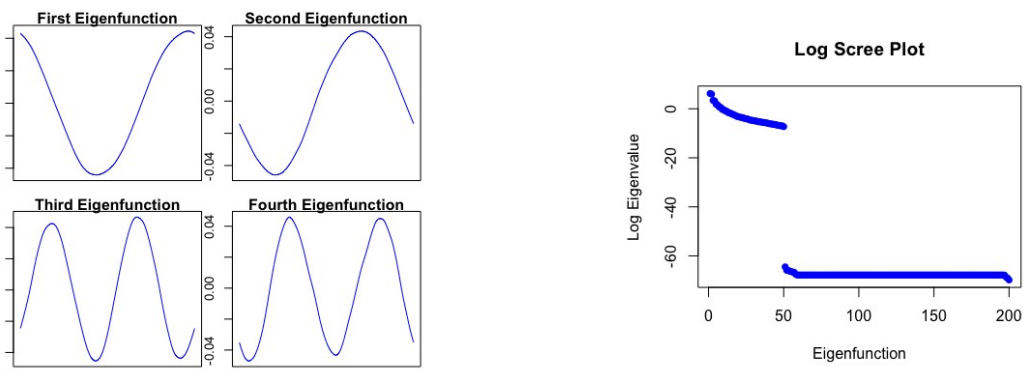


Figure 2: 200 random curves generated by an expansion of the Fourier basis.



(a) (b)

Figure 3: (a) The first four eigenfunctions returned by FPCA. (b) The scree plot (logarithm of eigenvalues for convenience).

Next, we investigate FPCA for sparse and irregularly sampled data. We return to sampling functions from (4.1), except now we only obtain 10 samples from each f_i at randomly chosen times (varying with each i). As explained in the methods section, a naïve implementation of FPCA will not work due to noise in μ and K. We presented the PACE algorithm as a solution to the problem of sparse and irregular sampling. An implementation of PACE is available in the R package `fdapace`, which is available for download on CRAN [4]. The results of FPCA with PACE are shown in figure 4. The mean function is quite noisy despite smoothing, which is indicative of the volatility in irregular sampling. The eigenfunction estimates align quite closely with the densely sampled case, although noise has clearly degraded these estimates to some degree.

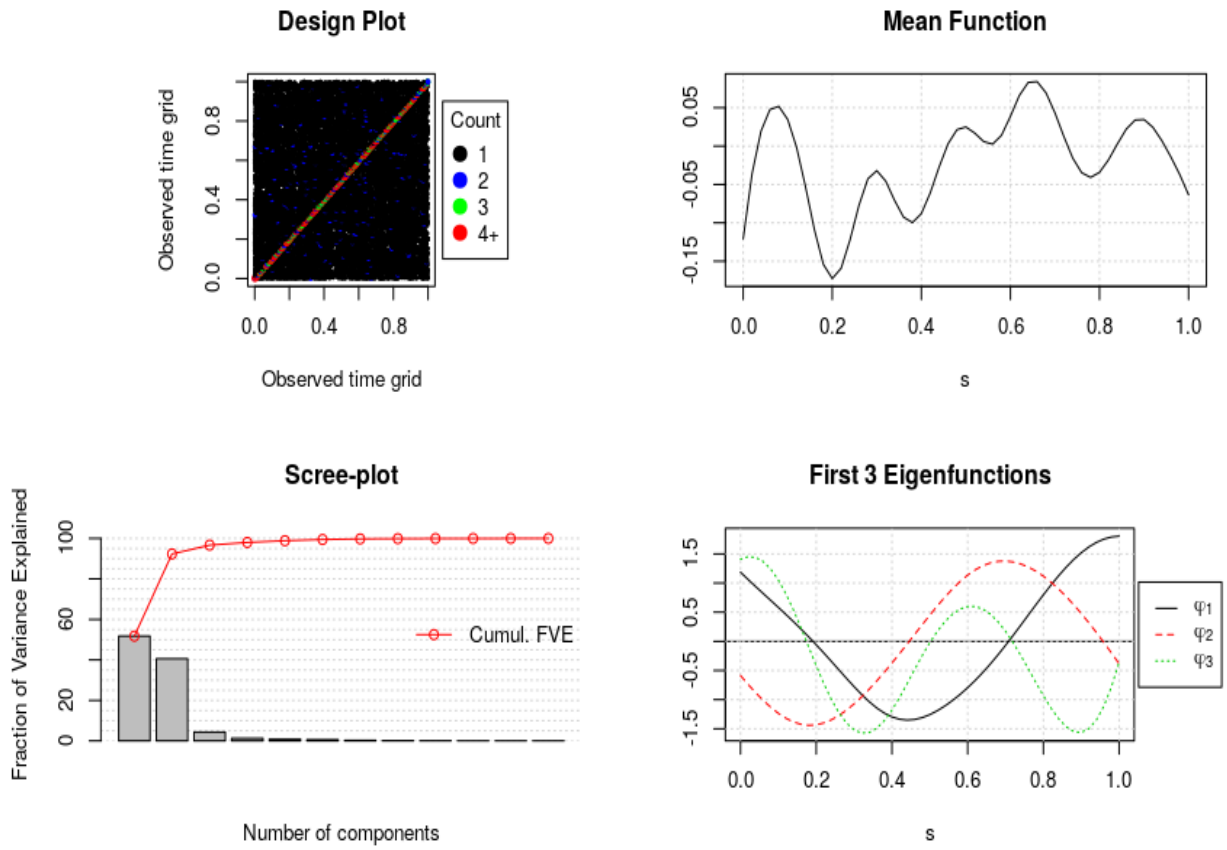


Figure 4: The results of FPCA for the sparse and irregularly sampled data from. This plot was obtained by calling the `FPCA` method from `fdapace`. The design plot counts the collection of all sampled time point pairs (t_i, t_j). A dense design plot indicates that there is enough data to get an accurate representation of the process (as the time points from individual functions are pooled together).

4.2 US Stock Market Data

FPCA is ideal for analyzing temporal data from financial markets. To demonstrate this, we obtain the opening prices of 6000 US stocks throughout the previous 365 days. The data was obtained from a free online database available at `https://stooq.com/db/h/`. We are interested in investigating how stocks prices in different sectors change throughout the year. FPCA cannot be immediately applied as the prices of stocks can be on wildly different scales. For example, a share of Amazon can cost well over \$1500 while other stocks are worth less than a dollar. Since we are only interested in the dynamics of the price curve, we divide each stock price by the maximum price that the stock achieved in the year, thus obtaining an adjusted price in the range $[0, 1]$.

Since the data is densely and regularly sampled, we can apply the standard FPCA algorithm. To make the estimated mean trajectory more interpretable, we apply locally weighted regression to the raw mean estimate. As shown in figure 5, stock prices took a major dip around 150 days after December 2018 (roughly May 2019).

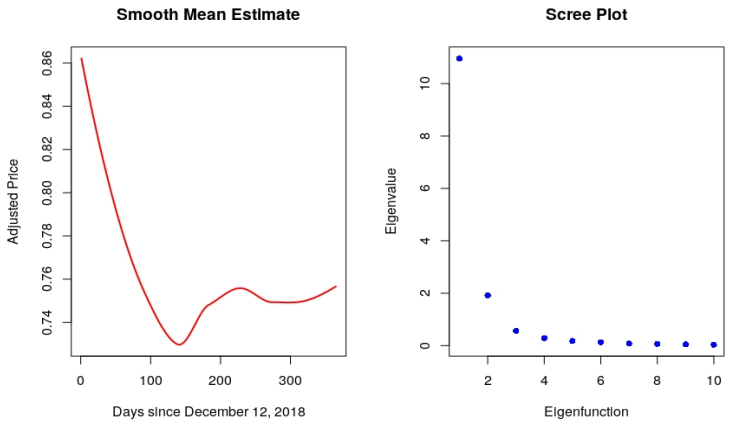


Figure 5: (Left) Smoothed estimate of mean trajectory. (Right) Scree plot for the first 10 eigenfunctions

The scree plot in figure 5 (right) indicates that the first few eigenfunctions can explain most of the variance in the data. The first four eigenfunctions, which were also smoothed to be more interpretable, can be found in figure 6. The first two eigenfunctions are essentially linear with a gentle slope. However, the eigenfunctions might not have a good physical interpretation as price is always non-negative and we have conditioned our adjusted price to hit 1 at some time point.

Finally, we compute the FPC scores using numerical integration. The plot of all 6000 stocks is not particularly informative as there are many small stocks that exhibit erratic behavior. These stocks usually correspond to extreme FPC scores, leaving all other stocks in one dense cluster. To simplify the picture, we focus only on major companies in a variety of different sectors. The following stocks (with tickers) were selected. **Technology:** Apple (AAPL), Facebook (FB), Microsoft (MSFT), Google (GOOGL). **Auto:** Ford (F), Chrystler (FCAU), General Motors (GM). **Food & Restaurant:** Pepsi (PEP), General Mills (GIS), McDonald's (MCD), Wendy's (WEN). **Transportation:** Norfolk Southern (NSC), CSX (CSX). **Energy:** ExxonMobil (XOM), BP (BP). The FPC scores for these stocks can be found in figure 7. In general, stocks in the same sector tend to be close together. Most notably, ExxonMobil and BP are very close together as well as 3 of the Food & Restaurant companies. Interestingly, technology appears to be the sector with the most separation between companies. A possible explanation for this is that the technology industry is less sensitive to external factors.

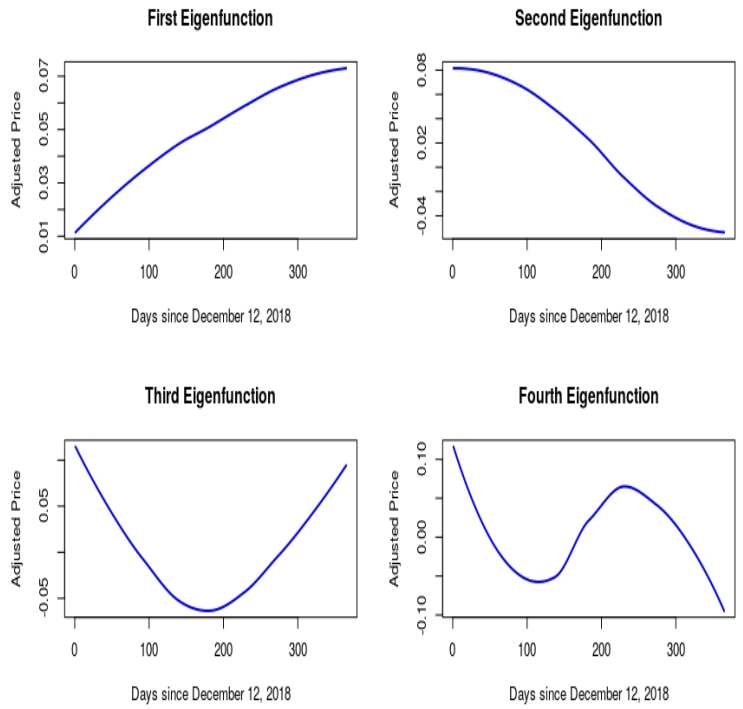


Figure 6: First four eigenfunction estimates. Locally weighted regression was applied to smooth the estimates

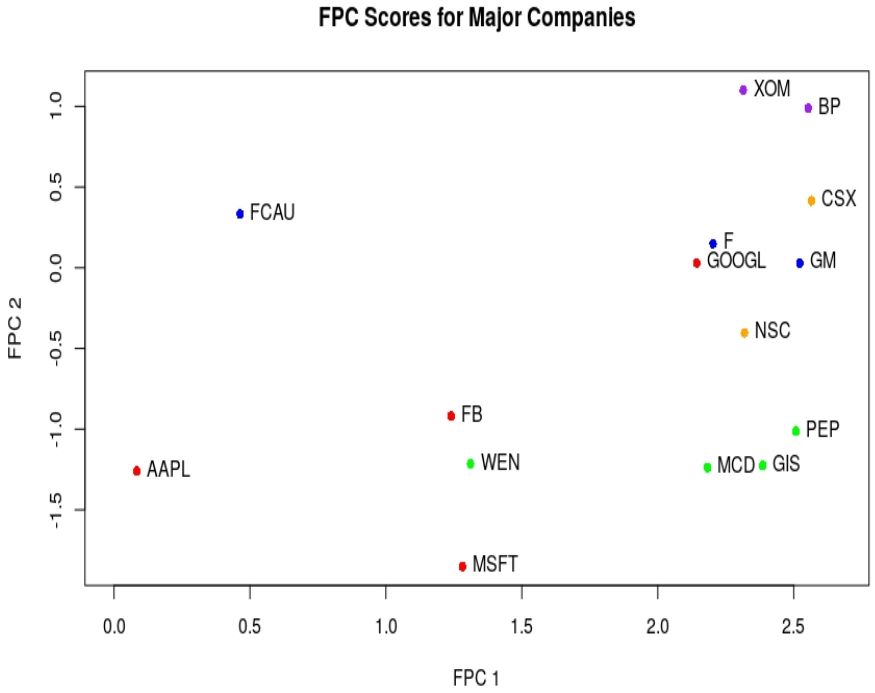


Figure 7: First two FPC scores for stock prices of major companies. Points are colored by sector.

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Appendix

A. Proof of Theorem 3

The proof is adapted from [6] (section 2.3.2). Denote the mean squared error of the approximation as ε_k. Then

$$\varepsilon_k = \mathbb{E}\left(\sum_{i=k+1}^{\infty} \langle X, \psi_i \rangle \psi_i\right)^2 \tag{4.2}$$

Expanding (4.2), we have

$$\begin{aligned}
\varepsilon_k &= \mathbb{E}\left(\sum_{i=k+1}^{\infty}\sum_{j=k+1}^{\infty} \langle X, \psi_i \rangle \langle X, \psi_j \rangle \psi_i \psi_j\right) && (4.3)\\
&= \sum_{i=k+1}^{\infty}\sum_{j=k+1}^{\infty} \mathbb{E}\langle X, \psi_i \rangle \langle X, \psi_j \rangle \psi_i \psi_j && (4.4)\\
&= \sum_{i=k+1}^{\infty}\sum_{j=k+1}^{\infty} \mathbb{E}\left(\int_{[0,1]}\int_{[0,1]} X_t X_s \psi_i(t)\psi_j(s)dtds\right)\psi_i\psi_j && (4.5)\\
&= \sum_{i=k+1}^{\infty}\sum_{j=k+1}^{\infty} \left(\int_{[0,1]}\int_{[0,1]} K(s,t)\psi_i(t)\psi_j(s)dtds\right)\psi_i\psi_j && (4.6)
\end{aligned}$$

Integrating ε_k over $[0,1]$, the $\psi_i\psi_j$ vanish for $i \neq j$ by orthogonality.

$$\int_{[0,1]} \varepsilon_k = \sum_{i=k+1}^{\infty}\left(\int_{[0,1]}\int_{[0,1]} K(s,t)\psi_i(t)\psi_i(s)dtds\right) \tag{4.7}$$

We want to minimize the error term subject to the constraint of orthonormality of ψ, so we can introduce lagrange multipliers λ_i and minimize

$$\sum_{i=k+1}^{\infty}\left(\int_{[0,1]}\int_{[0,1]} K(s,t)\psi_i(t)\psi_i(s)dtds\right) - \lambda_i\left(\int_{[0,1]} \psi_i^2(t)dt - 1\right) \tag{4.8}$$

Differentiating this functional equation with respect to ψ_i, we obtain

$$\int_{[0,1]} \int_{[0,1]} 2K(s,t)\psi_i(t)dtds - 2\lambda_i \int_{[0,1]} \psi_i(t)dt \tag{4.9}$$

Collecting terms and setting equal to zero, we have

$$\int_{[0,1]} \left(\int_{[0,1]} K(s,t)\psi_i(t) - \lambda_i \psi_i(t)dt \right) ds = 0 \tag{4.10}$$

This is minimized when

$$\int_{[0,1]} K(s,t)\psi_i(t) = \lambda_i \psi_i(t) \tag{4.11}$$

In other words, when ψ_i is the eigenfunction of K corresponding to the i-th largest eigenvalue. $\square$